

CHEM 101 Topic 2: Fundamental Measurements — Problem Set

Instructions: Complete the following problems using the dimensional analysis method practiced in your guided assignment. Show all your work, including given values, conversion factors, and units, to receive full credit. Round your final answers to the correct number of significant figures.

Section 1: The Metric System & Dimensional Analysis

Problem 1: A physician orders 0.75 g of a medication. How many milligrams (mg) should be administered?

Problem 2: A patient's IV bag contains 500 mL of normal saline. Convert this volume to Liters (L).

Problem 3: A laboratory measurement is recorded as 250 mcg (micrograms). Convert this value to milligrams (mg).

Section 2: Scientific Notation

Problem 4: The distance between two particles is 0.00000045 meters. Write this distance in scientific notation.

Problem 5: A chemical sample has a mass of 3.2×10^{-3} g. Write this mass in standard decimal form.

Section 3: Significant Figures

Problem 6: Determine the number of significant figures in the following measurements:

- a) 0.0506 g
- b) 1,200 mL
- c) 0.0010 L

Problem 7: Calculate the total mass of three combined samples: 5.22 g, 1.5 g, and 0.125 g. Round your answer to the correct number of significant figures.

Problem 8: Calculate the volume of a rectangular container with dimensions 1.2 cm x 3.45 cm x 2.0 cm. Round your answer to the correct number of significant figures.

Section 4: Temperature & Density

Problem 9: A patient's temperature is recorded as 38.5°C. Convert this temperature to both Fahrenheit (°F) and Kelvin (K).

Problem 10: A 20.0 mL sample of a liquid has a mass of 21.6 g. Calculate the density of the liquid and determine if it is more or less dense than pure water (1.00 g/mL).